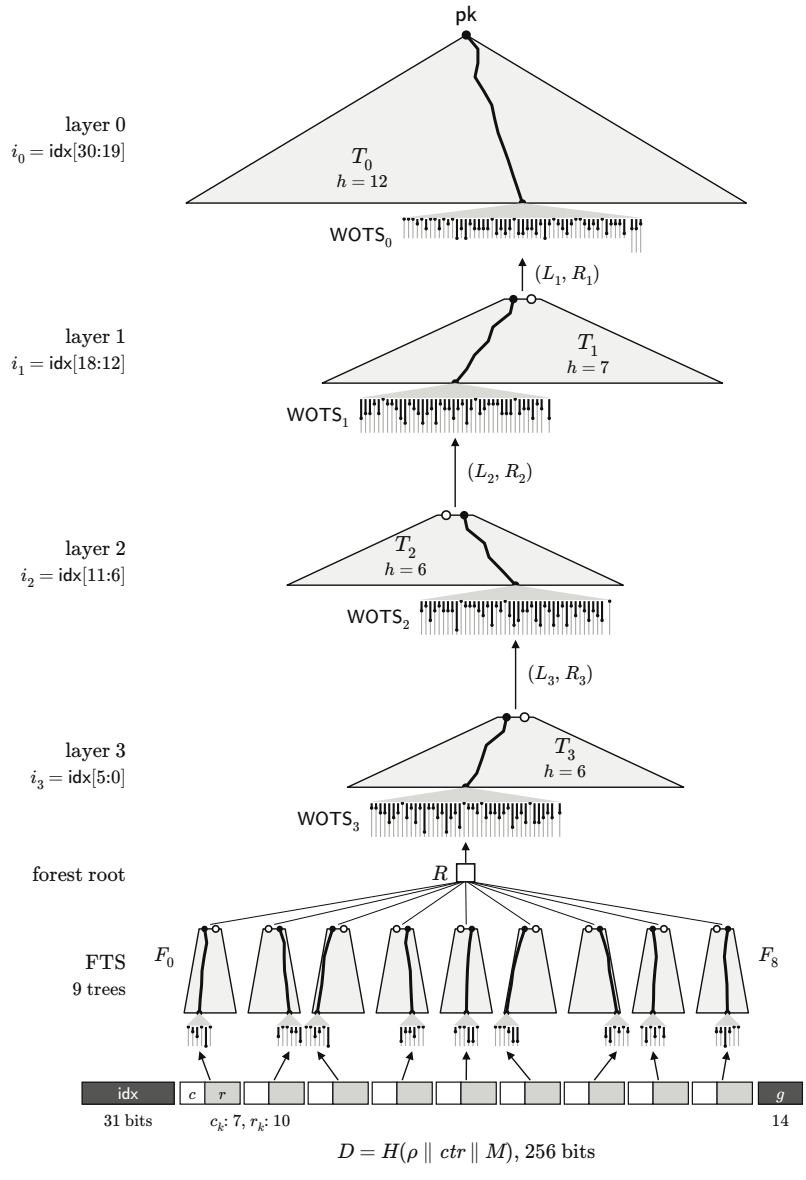


part	*bytes in σ*
$h = 12$: 4096 leaves, 1 tree, cached at key generation WOTS_0: 51 chains of 4 steps + 3 of 7; positions sum to 144 σ: 54 nodes + 12 auth. path	1,056
$h = 7$: 128 leaves, 2^{12} trees WOTS: 42 + 1 checksum chains of 7 steps; positions sum to 199 σ: 43 nodes + 7 auth. path	800
$h = 6$: 64 leaves, 2^{19} trees WOTS as in layer 1 σ: 43 nodes + 6 auth. path	784
$h = 6$: 64 leaves, 2^{25} trees WOTS_3 signs the forest root R σ: 43 nodes + 6 auth. path	784
$R = H(\mathsf{idx}, (L_k, R_k)_{k<9})$	
$h = 7$: 128 children per tree child: 6 chains of 4 steps; D picks child c_k and word w_k, $\Sigma\, d_i = 7$ σ: 9 × (6 nodes + 7 auth. path)	1,872
D valid iff $g < 1451$, all $r_k < 563$ *ctr* and layer counters not in σ	
randomizer ρ	16
$\lvert\sigma\rvert =$	**5,312**

end → hashed into the leaf
revealed in σ (position x_i)
seed

one chain, 7 steps
black: the verifier hashes $7 - x_i$ steps
grey: known only to the signer
FTS chains reveal position $4 - d_i$

● recomputed by the verifier
○ read from σ
□ one hash of many nodes